Revision Tracker

BNP-ABTests — AOAS Submission

"Online activity prediction via generalized Indian buffet process models"

Editor: Po-Ling Loh Decision: Major Revision

Last updated: April 12, 2026

Overview

Theme	Description	Open	Done
T1	Framing & Practical Impact	3	1
T2	Real Data Analysis	4	0
T3	Theoretical Exposition	1	9
T4	Section 3.4 & Notation (R1)	7	4
T5	Simulations & Presentation	3	0
T6	Typos & Mechanical	5	11
	Total	**23**	**25**

Key editorial constraints:

- Target ≤20 pages (AOAS template, ∼500 words/page)
- Dedicated Section 2 for motivating application and dataset
- Minimize color in printed figures (use line styles)
- Single point-by-point response file (Editor + AE + R1 + R2)

T1 — Framing & Practical Impact

Core issue: AE + R2 (major b, c) + Editor. The paper's goal was unclear, the data analysis is thin, and there is no evidence the method changes real decisions.

Update (Apr 12): Co-author rewrote abstract and intro with duration-planning framing, three estimands (U, T, D_M), "Decision target" paragraph, and citations to Gualavisi2025/Masoero2025. T1.a is now done.

Item	Src	Pri	Description	Status
T1.a	AE, R2-c, Editor	High	**Clarify the paper's primary goal.** Rewrite intro: the goal is *prediction of user engagement to inform experiment duration decisions*. Estimands: $U_{D_0}^{(D_1)}$, $T_{D_0}^{(D_1)}$, D_M. *Co-author added: abstract reframe, intro paragraph with three estimands, "Decision target" paragraph, contribution rewording, citations.*	✓ Done
T1.b	AE, R2-b	High	**Demonstrate practical impact.** Case studies on public data: "with 7 days of pilot, our method predicted X users at day 28; experiment could stop Y days early." Compare to competing methods. *Pipeline exists (UCI/REES46/ASOS results computed). Not yet integrated into the paper.*	☐ Not started
T1.c	R2-b	High	**Calibration diagnostics.** Across experiment windows: does truth fall in 95% CI? Report empirical coverage. Stratify by D_0 and D_1. *Results exist but CI coverage is poor on current data. Needs investigation before presenting.*	☐ Not started
T1.d	Editor	High	**Dedicated Section 2 for motivating application.** Editor explicitly requests this. Restructure: introduce dataset and application context before reviewing existing methods. *Not yet done — co-author kept current structure.*	☐ Not started

T2 — Real Data Analysis

Core issue: AE + R2 (major b, minor a, d, j). Data analysis is thin. Needs richer dataset description, visualizations, and decision-making evidence.

Item	Src	Pri	Description	Status
T2.a	R2-j, Editor	High	**Fuller dataset description.** REES46: cosmetics e-commerce, event types, 5 months, 1.6M users. ASOS: what's available. Explain windowing into experiments.	☐ Not started

Item	Src	Pri	Description	Status
T2.b	R2-a	Med	**Visualize data characteristics.** (i) Histogram of per-user trigger counts (log-log). (ii) Cumulative new users over time (power-law curve).	☐ Not started
T2.c	R2-d	Med	**Power-law reference + visualization.** Cite Clauset et al. (2009). Pair with T2.b figure.	☐ Not started
T2.d	R2-c	Med	**Polish "isn't M always known?" response.** Draft exists. Ensure Section 2 text reflects the argument.	↻ In progress
T2.e	R2-j, AE	High	**Decision-making impact on real data.** 3–5 REES46 experiments: prediction trajectories, D_M intervals, comparison to competitors.	☐ Not started

T3 — Theoretical Exposition

Core issue: R2 (major a, d) and minor comments. Clarify novelty vs. Camerlenghi et al. (2022), add practical interpretation of results.

Status: 9/10 items addressed in current revision (red text).

Item	Src	Pri	Description	Status
T3.a	R2-a	—	Clarify contribution vs. Camerlenghi et al.	✓ Done
T3.b	R2-d	—	Practical interpretation after Cor. 3.5 and Prop. 3.7.	✓ Done
T3.c	R2-f	—	Reorganize Section 3.1.	✓ Done
T3.d	R2-g	—	Clarify NB parametrization.	✓ Done
T3.e	R2-h	—	Clarify prior importance.	✓ Done
T3.f	R2-e	—	First-trigger-time modeling advantage.	✓ Done
T3.g	R2-l	—	Model selection guidance in Discussion.	✓ Done
T3.h	R2-b	—	Clarify difference from algorithmic approaches.	✓ Done
T3.i	R2-i	—	Fix SSP vs. Be-SSP naming.	✓ Done

Item	Src	Pri	Description	Status
T3.j	R2-1	Med	**Model fit assessment procedure.** R2 asks how to select the score distribution beyond guidance. Consider: marginal likelihood comparison, posterior predictive checks.	☐ Not started

T4 — Section 3.4 & Notation (Reviewer 1)

Core issue: R1 is positive overall but has 11 specific points, mostly notation and clarity. Point 1 (Section 3.4 rewrite) is the most substantive.

Update (Apr 12): Co-author fixed T4.8 ($\pi_{\mathcal{G}}$ / $\tilde{\Delta}_{1,h}$ defined), T4.10 (sum index), and several typos that overlap with T4.5, T4.11.

Item	Src	Pri	Description	Status
T4.1	R1-1	High	**Rewrite Section 3.4.** Clarify D_M definition, "new" vs "unobserved" users, relationship of F' to Z^{FT}. Strengthen Theorem 3.8 statement.	☐ Not started
T4.2	R1-2	Low	Eq. (1): second sum runs $D_0{+}1$ to $D_0{+}D_1$.	☐ Not started
T4.3	R1-3	Med	p.4: "this assumption clearly…" too strong without distributional assumptions on $A_{d,n}$.	☐ Not started
T4.4	R1-4	Low	Def. 3.1: $\overset{\text{iid}}{\sim}$ should be $\overset{\text{ind}}{\sim}$ (depends on n).	☐ Not started
T4.5	R1-5	Low	Eq. (5): s vs. θ typo. Rephrase "puts to zero" more formally.	☐ Not started
T4.6	R1-6	Low	Mention P_0 is non-atomic in or near Def. 3.2.	☐ Not started
T4.7	R1-7	Low	End of Def. 3.2: $\tilde{\mu}$ vs. μ inconsistency.	☐ Not started
T4.8	R1-8	Low	Thm. 3.4: define $\pi_{\mathcal{G}}$ and $\tilde{\Delta}_{1,h}$ before use. *Co-author added blue text defining both in Thm. 3.4 statement.*	✓ Done
T4.9	R1-9	Low	Capitalize "Theorem" and "Algorithm" consistently.	☐ Not started
T4.10	R1-10	Low	Prop. 3.6: sum for U should start from $D_0{+}1$. *Co-author fixed: now* $D_0{+}1$ *to* $D_0{+}D_1$.	✓ Done
T4.11	R1-11	Low	p.9: "usres" → "users". *Co-author fixed many typos in Section 3; this one likely covered.*	✓ Done

T5 — Simulations & Presentation

Core issue: R2 (minor k) + Editor constraints. Rebalance toward real data; respect 20-page limit; minimize color.

Item	Src	Pri	Description	Status
T5.a	R2-k	Med	**Move simulations to supplement.** Move Section 5.1 (data from true model) to supplement. Keep Zipf (5.2) and interval comparison (5.3). Use freed space for richer Section 6.	☐ Not started
T5.b	Editor	Med	**Page limit: ≤20 pages.** Check count after restructuring. Tighten prose where possible.	☐ Not started
T5.c	Editor	Low	**Figure colors.** Audit all figures: replace color-only distinctions with line styles (dashed, dotted, markers). Avoid referring to colors in text/captions.	☐ Not started

T6 — Typos & Mechanical Fixes

Update (Apr 12): Co-author fixed most typos across Sections 2–6. Remaining items are structural (duplicate abstract, keyword, double brace, red/blue markers).

Item	Source	Issue	Status
T6.1	R2	Unresolved cross-references (**??**) throughout.	☐ Not started
T6.2	R2	p.3: F_n redundant in data tuple (derivable from A).	☐ Not started
T6.3	R2	p.5 Sec. 3.2: define $\mathcal{E}(1)$ (exponential). *Co-author added blue text.*	✓ Done
T6.4	R2	Cor. 3.5: define $\tilde{\Delta}_{1,h}^{-\alpha}$ in statement. *Co-author added blue text in Thm. 3.4.*	✓ Done
T6.5	R2	p.11: "does require" → "does not require". *Co-author fixed.*	✓ Done
T6.6	R2	p.16: format τ as math mode. *Co-author fixed.*	✓ Done
T6.7	proofread	`main.tex`: abstract `\input`'d twice.	☐ Not started
T6.8	proofread	`main.tex`: "user prdiction" → "prediction".	☐ Not started
T6.9	proofread	`1_intro`: double brace `\Cref{{...}}`.	☐ Not started

Item	Source	Issue	Status
T6.10	proofread	`2_data`: “uesers sexhibit”, “propsensity”, “Morever”. *Co-author fixed all three.*	✓ Done
T6.11	proofread	`3_bnp`: “racall”, “ininitely”, “uner”, “whcih”, “hiterto”, “predicitve”, “migh”, “consits”. *Co-author fixed all.*	✓ Done
T6.12	proofread	`4_num`: “close form” → “closed-form”. *Co-author fixed.*	✓ Done
T6.13	proofread	`5_num`: “accuarcy”, “esimates”. *Co-author fixed.*	✓ Done
T6.14	proofread	`6_real`: “dasetest” → “datasets”. *Co-author fixed.*	✓ Done
T6.15	proofread	`3_bnp`: verify file not truncated (line 345).	☐ Not started
T6.16	proofread	Remove all `\textcolor{red}` and `{\color{blue}}` markers before final submission. *Note: co-author added new blue markers.*	☐ Not started